

SIDHARTHA PULLURI

Richardson, TX | 214-475-4851 | Sidhartha.a.1729@gmail.com | LinkedIn | *Seeking Summer 2026 Internship*

Education

The University of Texas at Dallas

Master of Science in Computer Science

Richardson, TX

Aug 2025 – Dec 2027

- Relevant Coursework: Natural Language Processing, Artificial Intelligence, Database Design, Advanced Machine Learning

Chaithanya Bharathi Institute of Technology

Bachelor of Science in Computer Science | GPA: 8.7/10.0 (First Class with Distinction)

Hyderabad, India

Aug 2021 – May 2025

- Relevant Coursework: Machine Learning, Computer Vision, Deep Learning, Natural Language Processing, Data Structures

Technical Skills

Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, Hugging Face Transformers, LangChain

Computer Vision & NLP: OpenCV, NLTK, spaCy, YOLO, EfficientDet, Vision Transformers, BERT, GPT

Programming: Python (NumPy, Pandas, Matplotlib), SQL, R, JavaScript, C++

Tools & Platforms: Git, Docker, Jupyter, Google Colab, AWS, Flask, FastAPI, Gradio, Weights & Biases

Data Management: MySQL, PostgreSQL, SQLite, Vector Databases (Pinecone, Chroma)

Machine Learning Projects

AI-Powered Audio Storytelling System | *Python, Bark TTS, NLP*

Aug 2025 – Present

- Architected end-to-end audio production pipeline for long-form narrative generation, processing 50+ chapters with automated text segmentation, punctuation normalization, and voice modulation
- Optimized TTS inference pipeline using batch processing and GPU acceleration, reducing audio generation latency by 40% while preserving natural prosody across 10+ hour audiobooks
- Engineered chapter-wise audio assembly system with dynamic silence padding and seamless transitions, validated through beta testing with 200+ users achieving 95% satisfaction
- Implemented automated quality control checks for audio artifacts, clipping detection, and volume normalization across production pipeline

Medicinal Leaf Recognition System | *PyTorch, EfficientNet, BiFPN, Gradio*

Jul 2024 – Mar 2025

- Designed custom CNN architecture combining EfficientNetB0 backbone with Bidirectional Feature Pyramid Network (BiFPN) for multi-scale feature extraction, achieving 94.2% top-1 accuracy on 10,000+ image dataset of Indian medicinal plants
- Implemented comprehensive data augmentation pipeline (geometric transforms, color jittering, mixup) improving model robustness by 12% on real-world mobile captures with varying lighting and backgrounds
- Deployed production-ready web application using Gradio with REST API integration, serving 500+ daily predictions with <1s inference time and integrated Ayurvedic medicinal information database
- **Publication:** First-author paper presented at IEEE-affiliated International Conference on Innovative Computing Technologies (ICICT-2024), Hyderabad; contributed model to open-source medicinal plant recognition community

GPT-Powered Conversational AI Chatbot | *Hugging Face, Transformers, Flask, Redis*

Feb 2024 – Jun 2024

- Built production-grade chatbot using GPT-2 (774M parameters) with custom fine-tuning on domain-specific corpus of 50K+ dialogues, reducing perplexity score by 28% and improving contextual relevance
- Designed secure token-based authentication system (JWT) and implemented stateful conversation management using Redis for multi-turn dialogues with context retention across 10+ exchanges
- Deployed scalable REST API with Flask and load balancing achieving 99.5% uptime, handling 1,000+ concurrent users with automated performance monitoring, logging, and real-time analytics dashboards
- Integrated response caching and rate limiting mechanisms reducing average API response time from 2.1s to 0.8s

Technical Leadership & Recognition

CBIT Hackathon 2024, 3rd Place: Led cross-functional team of 4 to develop ML-powered health diagnostics tool using Random Forest and XGBoost models, competing against 50+ teams over 48-hour sprint

Published Researcher: First-author publication on efficient object detection architectures for resource-constrained devices at IEEE-affiliated ICICT-2024 conference

Placement Coordinator: Managed end-to-end technical recruitment process for 200+ CS students, coordinating placement drives with 30+ top-tier tech companies including Google, Microsoft, Amazon, and Meta (Jun 2024 – May 2025)

Active ML Community Member: AI BUZ Club (UTD), Artificial Intelligence Society (UTD), Neural Nexus Club (CBIT 2023-2025)

Certifications

Machine Learning and Image Processing – L&T Technologies & NASSCOM Foundation (Mar 2024)

Machine Learning Specialization – Internshala (Jun 2023)